Problems with software projects

Problems related to project management

- unclear objectives
- Poor planning
- project management methodology correctly not followed
- Changing to new technology
- insufficient staff

- Effective project management is key to successfully executing a project

Problems with software projects contd..

- Problems with project planning

Lack of Leadership

- A talented and motivated team who are involved throughout the project can make a successful project.

Deviation from scope

- A change in project priorities due to lack of understanding of the scope of the project
- When customer needs are not well understood, wrong software is built

Communication gap

- There is a communication gap between the people who build the software and those who need it.
- You can mitigate this problem by letting people know the current progress on a regular basis and when milestones are accomplished or the status changes.

Fixing Planning Problems

- Use a vision and scope document to define the needs of the users and stakeholders
- Use a project plan to keep every one informed about how those needs will be met
- Use risk planning to keep the plan realistic

Problems with Project estimates and scheduling

- A project manager under pressure simply imposes a deadline, and creates unrealistic estimates that meet it

  - **Unrealistic estimates** shorten the tasks in the schedule or cutting them entirely until everything fits

  – When the schedule gets tight, any non-programming activities are cut and the software is released before it’s finished

Problems with Project estimates and scheduling contd..

- **Disconnected business priorities**

  There is often a wide gap between the priorities of projects within the development, organization and the priority of the project in the eyes of the requester.

  High priority project for one department is not viewed as important by the organization as many projects may be there for the organization

  Everyone needs to be the same with respect to Priorities

Problems with Project estimates and scheduling contd..

- **Adding people to a project**

  It may take only a few days or weeks for a developer to come into the project at the beginning, but it could take months for a developer to be fully productive when added to the project long after it has started

- Multi-tasking

  The more you ask people to multi-task, the worse their work will be and the longer it will take.

Fixing Scheduling Problems

- These can be avoided by adopting good planning and estimation practices and creating a project schedule
- Scheduling techniques like critical path analysis can help spot problems early on

Review Problems

Problems Are Found Too Late

– The team may misunderstand a need, but that's not discovered until delivery

Requirements may be missed or incorrect

– The design may be difficult to use or fail to take all of the features into account

Review Problems contd..

Long Meetings

– To solve some persisting problem the project manager conducts long meetings; the meetings may end up without making any progress

Non availability of Indispensable persons

– Important persons like senior programmer may not be available to discuss the bugs

Fixing Review Problems

- Problems that are found too late, big meetings, and the indispensable persons like programmer are problems which can be solved with reviews
  - Reviews can catch defects early, when they are cheaper to fix
  - A review meeting only includes the people necessary for the work to be done
  - Reviews especially code reviews can help the senior programmer spread his expertise and knowledge

Problems with software projects

- People-related problems
- Process-related problems
- Product-related problems
- Technology-related problems

People-related problems

Low motivation: The project manager is responsible for the level of motivation within the team.

Lengthy projects, complex activities and scarce resources often decrease the motivation level in a software development team

- The project manager has to constantly motivate the people to do a good job.

People related problems contd..

- **Problem employees**: Some members of any team always create a problem. They lower the efficiency and productivity of other team members and make it difficult to meet the objectives of the software project within the specified time.
- The project Manager has to ensure that employees are not allowed to create a problem for the rest of the team, even if the employee is very competent

People related problems contd..

- **Unproductive work environment:** The work environment is a major factor that affects the productivity of the development team; for example, a noisy or cramped workspace decreases the motivation levels of the employees.

  Similarly, unfriendly organizational policies also lower the motivation of the team members.

  The project manager needs to ensure that the team is

  protected from such external disturbances and make the workspace friendly to work in.

People related problems contd..

- **Inefficient project management style*:*** the project manager needs to lead by example
- *The team members absorb the work culture*, work ethic, and attitude of the project manager and implement it in their work style.
- If he displays a lack of leadership qualities and weak ideals, the motivation levels decrease across software team.

People related problems contd..

- **Lack of stakeholder interest:** For a software project to be a success, each stakeholder needs to take an active interest in the progress of the project.
- All stakeholders, including the customer, the management, and the software development team, need to commit to the success of the project.
- If the software development team is not committed to the project, then their contribution may not be to the optimum level.

People related problems contd..

Ineffective project sponsorship by management**:** Lack of commitment of the senior management to a software project lowers the motivation level of the team members.

- If the management commits to the progress of a software project, and takes a keen interest in the progress, the confidence of the software development team will increase.

Process- related Problems

Unrealistic schedule: Assigning unrealistic deadlines for a software project is

a primary reason why software projects are delayed. Often, the marketing or the management team commit a delivery date to the customer in the hope of getting the project contract.

- However, these dates many times are not decided in consultation with the development team.

- It is essential to ensure that the deadlines match the ability of the software team to deliver the software product.

- As it is not always possible to shift deadlines committed to the customer, you also need to plan the resource allocation and project execution such that the deadlines are met.

Process- related Problems contd..

- **Abandoning quality under pressure of deadlines:** Where a software project faces a shortage of resources, time, and funds,

  project managers often push away quality concerns and focus on meeting deadlines and staying within the budget.
- Abandoning quality adds even more time, effort, and costs to the software projects.
- The cost of doing things right the first time is lower than the cost of inspection during product delivery.
- Also, the cost of inspection is lower than the cost of debugging software after the customer spots errors.

Process- related Problems contd..

- **Unstructured and hurried software development:**

  When software project progresses with more focus on meeting deadlines and staying within a budget,

  the approach to the software development is unstructured and hurried.

- One should plan the software project such that all the activities are identified, sequenced properly, and roles and responsibilities assigned to the various people on the project

- One should also maintain the focus of the development team toward a structured approach to software development.

Process- related Problems contd..

- ***Insufficient Risk identification:*** Unidentified, partially identified, and unplanned

  risks pose a threat to the success of a software project.

  It is required to intensively identify risks and evolve a risk management plan such that the project is

  completed successfully, on time

Process- related Problems contd..

- **Unsuitable life cycle model selection**: Different software projects require different SDLC models. For example, a project to create banking software is different from

  software for a satellite where the concept needs to be researched
- For the former example, the Waterfall model is more applicable
- For the latter example, the Spiral model is more suitable.
- Selecting the correct life cycle model is critical to the success of a software project.

product-related problems

- **Product scope changed towards the end of the project life cycle:** The project time, effort, and cost estimates for a software project can go up dramatically when the customer changes the scope of the product toward the end of the project.
- In such situations, the criticality of the scope change should be verified
- If the change request is not critical, the original scope can be retained with a proper explanation to the customer.
- If the change request is critical, explain the situation to the customer.
- Usually, a customer gives more time and funds to a software project if proper justification is provided.
- In some cases, the scope change may also be because of a change in government policy; it may become mandatory for you to include such change requests.

Technology related problems

- **Research-oriented software development*:*** Many software projects deviate from the original scope because of the nature of the software product or technology used

- When a totally new kind of software is developed or a new technology is used, the software development team can lose focus of the objectives by getting into a research-oriented approach

- It becomes the responsibility of the project manager to maintain the focus on the objective

COST-BENEFIT ANALYSIS

Introduction

Feasibility study

- Operational feasibility is centered on human factors as is the problem worth solving ?
- What are the clients expectations ?

  Technical feasibility is concerned with respect to computer and hardware.
- Can the problem be solved on computer?
- Is the technology available?
- Is expertise to use technology available?

Feasibility Contd.

Economic feasibility

- costs and benefits
- Early estimates based on heuristics
- Detailed estimates and cost-benefit analysis

Clients and analysts must evaluate the results of all the three analyses and make decisions based on that

COCOMO (Constructive Cost Model) is a cost model

Putnam Model is a resource estimation/effort estimation model

Cost benefit analysis

- Cost benefit analysis is a systematic approach to estimate the economic feasibility of a system considering the incurred costs and estimating the return on investment

- It involves System costs, and benefits from the system.
- If the system is economically feasible means the benefits outweigh the system costs within a defined period of time acceptable to the user/client.

SYSTEM COSTS

1. Development costs: The following are some development costs

One time expense for the project	Furniture chairs etc.	Facility costs (prepare facility to operate the system, AC etc.)
Personnel costs (salaries) etc.	Computer usage costs	Miscellaneous expenses (rent, utilities etc.)
User time	Supply and materials costs	correspondence costs (Postal, shipping costs)
Training costs	Peripherals, office supplies consumed	
Equipment costs (hardware, software)	Costs of manual and documents	

System cost Contd…

2. Ongoing Operating costs: recurring costs for operation of the system, maintenance costs etc.

3. Fixed costs: occur at regular intervals as property taxes, lease payments

   Company provides some benefits to employees as hospital, market etc.

4. Variable costs : base on the usage of the system (CPU time, supply costs, wage costs, maintenance costs)

BENEFITS

Increase profits or decrease costs

1. Performance Benefits:

- To improve quality of work or to start new activities
- Error reduction, speed of activity, access to new information etc.

2. Cost-Avoidance Benefits:

- Some clerical staff can be reduced due to new system
- Manual errors can be less due to more automation

Benefits contd..

3. Tangible Benefits:

- Can be estimated accurately
- Measures as monthly or annual savings
- Staff reduction
- Manufacturing a new part

4. Intangible benefits:

- Goodwill
- Environmental friendly

Quantifying intangible benefits

- Method 1 : Subjective estimation
- User makes estimation based on experience
- Analysts document and add to cost-Benefit report

Method 2 : Decision Theory (based on expected values)

- Assign probabilities to various outcomes.

If better working conditions are provided, by what percentage people leaving company is reduced ? For eg. 50 % chance of <4 % reduction

If less people leave the organization, savings in terms of termination pay, recruitment cost and training costs etc.

Quantifying intangible benefits contd..

- Method 3 : Working backwards from desired results.

  Better working conditions cost Rs. 100 000

  Savings of a person not leaving Rs. 18 700

  Hence reduce (turnover of people) or people leaving by = ~5.35 people

  (100 000 / 18700 = ~5.35)

COST- BENEFIT TERMINOLOGY

- Spending vs. Investment: Spending is purchasing what is needed for the operation of the system. Investment is the money spent and it can generate revenue. Goal is to maximize revenue.
- Direct versus Indirect Costs and Benefits: Direct cost and benefits are related to the system

  Indirect costs and benefits are expenses and revenue that are shared with the other components of the organization.

  For eg. The director of an organization is involved in many projects.

COST- BENEFIT TERMINOLOGY Contd..

- Economic Life: Period of time the investment is expected to yield benefits.

  - *Residual value*: the value of parts of the existing system that can be salvaged (relieved).

  - *Required rate of Return*: Minimum rate of return a company can accept on an investment.

- Sunk costs: Costs related to an investment which have already been paid out

  Just because $ 20000 (sunk cost) has already been invested in a project, you cannot justify proceeding with the project.

Techniques Used to Determine Economic Feasibility

1. Payback Analysis
2. Return on Investment
3. Net Present Value
4. Profitability Index

The *required inputs* to apply the techniques are:

1. Development costs
2. Annual Operating Costs
3. Annual Benefits
4. Economic life of the system (5 yrs.)
5. Required rate of return (15%)

Techniques Used to Determine Economic Feasibility Contd..

1. Payback Analysis is to determine in how much time the accrued benefits overtake accrued costs. Breakeven point is when debts have been cancelled by benefits.

Payback period (T) = $\frac{\textit{Development costs (D)}}{\textit{Annual Revenue-annual operational costs (P)}}$

2. Internal Rate of Return (IRR): is a measure of relationship between investment in a project and net benefits from the project.

IRR = $\frac{\textit{estimated lifetime benefits – estimated lifetime costs}}{\textit{estimated life time costs}}$

To calculate yearly IRR, divide lifetime IRR by the economic life of the system.

Techniques Used to Determine Economic Feasibility Contd..

3. Present Value & Net Present Value (NPV)

   Costs and benefits of the system are calculated in terms of today's dollars (i.e. the value of the investment in today's dollars)

   NPV = *Present value of total benefits – present value of total costs*

4. Profitability Index (PI)

   Attempts to measure risk by determining ratio of present value of benefits to the development of costs.

$$PI = \frac{\text{present value of benefits}}{\text{present value of development costs}}$$

Stages of a project: Main activities in Project Management

- Feasibility study (Operational, Technical, and Financial)

  Is it worth doing?
- Planning (if feasible)

  How do we do it?
- Execution (implement the plan)

  Do it

Main activities in Project Management contd..

- Feasibility study is performed for large and complex projects

- Planning is done before the main engineering Life cycle (LC) and closure after the LC

- Monitoring phase is in parallel with LC

Project Planning

- Basic objective: To create a plan to meet the commitments of the project, i.e. create a path that, if followed, will lead to a successful project
- Planning involves defining the Life Cycle (LC) process to be followed, cost and time estimates, detailed schedule, plan for quality, etc.
- Main output - a project management plan and the project schedule

Key Planning Tasks

- Define suitable processes for executing the project
- Estimate effort
- Define project milestones and create a schedule
- Define quality objectives and a quality plan
- Identify risks and make plans to mitigate them
- Define measurement plan, project-tracking procedures, training plan, team organization, etc.

Process Planning

- Plan how the project will be executed, i.e. the process to be followed
- Process will decide the tasks, their ordering, milestones
- Hence process planning is an important project planning task
- Should plan for LC and PM processes as well as supporting processes

Life Cycle Process

- Various LC models - waterfall, iterative, prototyping; diff models suit different projects
- During planning can select the model that is best for the project
- This gives the overall process which has to be fine-tuned to suit the project needs
- Usually done by process tailoring - changing the process to suit the project
- Tailoring finally results in adding, deleting, modifying some process steps